

1. Dataset Generation

A synthetic cybersecurity dataset was created to simulate real-world scenarios, encompassing various metrics such as Timestamp, Source_IP, Packet_Size, Login_Attempts, File_Encryption_Rate, and Threat_Type. Intentional data quality issues, including missing values in Packet_Size and File_Encryption_Rate, and duplicate rows, were introduced to mimic common challenges in raw data.

2. Data Cleaning

To ensure data integrity and reliability for analysis, the following cleaning steps were performed:

Duplicate rows were identified and removed.

Missing values in Packet_Size were imputed using the median of the column.

Missing values in File_Encryption_Rate were filled with 0, assuming no encryption activity when data was absent.

Data types were verified and corrected, particularly converting the Timestamp column to a datetime format.

3. Exploratory Data Analysis (EDA)

Initial exploratory analysis revealed:

Distributions of numerical features (Packet_Size, Login_Attempts, File_Encryption_Rate) show varied patterns, providing a baseline understanding of their typical ranges.

Correlations between numerical features were generally low, suggesting they represent distinct aspects of system activity.

Threat Type Analysis: Box plots illustrated how metrics like File_Encryption_Rate and Login_Attempts vary across different Threat_Type categories. For example, 'Ransomware' and 'DDoS' showed different patterns compared to 'Clean' activities.

4. Statistical Analysis

To differentiate between malicious 'Threat' activities (Ransomware, DDoS) and 'Clean' system behavior, a detailed statistical analysis was conducted:

A new feature Is_Threat was created to categorize Threat_Type into 'Threat' or 'Clean'.

Group-wise statistics (mean, median, standard deviation) were calculated for Packet_Size, Login_Attempts, and File_Encryption_Rate across all Threat_Type categories.

Key Indicator Insights (Threat vs. Clean):

Ransomware: Showed a slightly higher average Packet_Size (+3.25%) but lower average Login_Attempts (-8.76%) and File_Encryption_Rate (-24.83%) compared to 'Clean' activities. The lower encryption rate for Ransomware is an interesting observation in this synthetic dataset, possibly indicating that the 'File_Encryption_Rate' feature primarily captures legitimate encryption processes, and its absence or reduction could be an indicator of a different kind of threat.

DDoS: Exhibited a notably lower average Packet_Size (-21.01%) and slightly lower Login_Attempts (-3.97%) and File_Encryption_Rate (-4.56%) compared to 'Clean' activities. The smaller packet size aligns with typical DDoS attack characteristics.

5. Feature Engineering

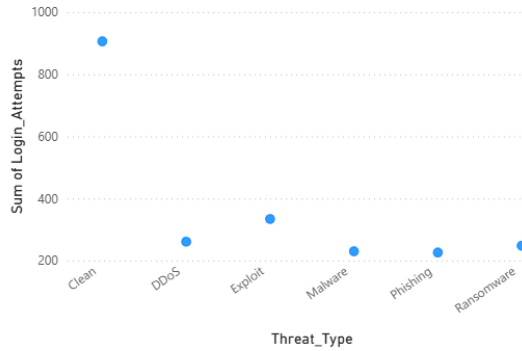
A new derived feature, Risk_Score, was engineered to combine the normalized Login_Attempts and File_Encryption_Rate into a single indicator of potential risk. This score was normalized to a 0-100 scale, aiming to provide a consolidated metric for threat detection.

6. Data Export

The cleaned and engineered dataset has been successfully exported to a CSV file (power_bi_export/cybersecurity_data.csv) for further visualization and interactive analysis using Power BI.

Visualisation using Power BI

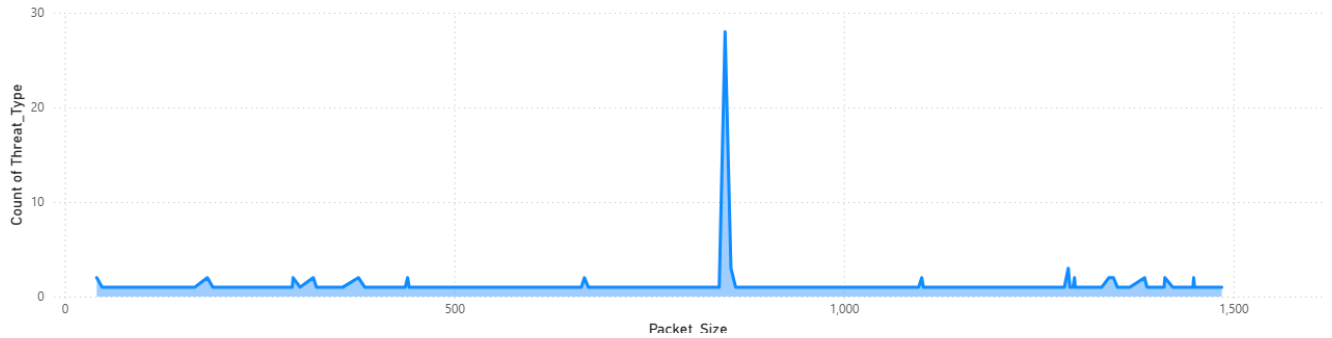
Sum of Login_Attempts by Threat_Type



Count of Threat_Type by Threat_Type

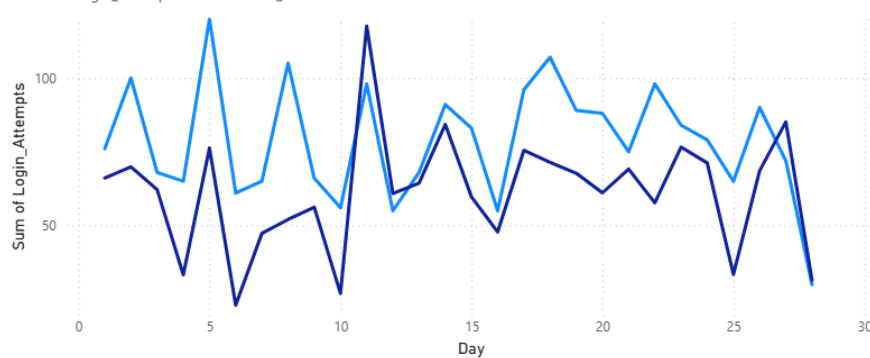


Count of Threat_Type by Packet_Size

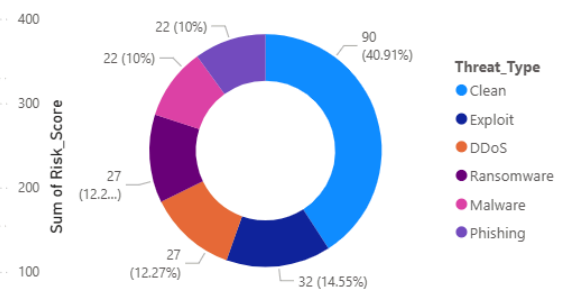


Sum of Login_Attempts and Sum of Risk_Score by Day

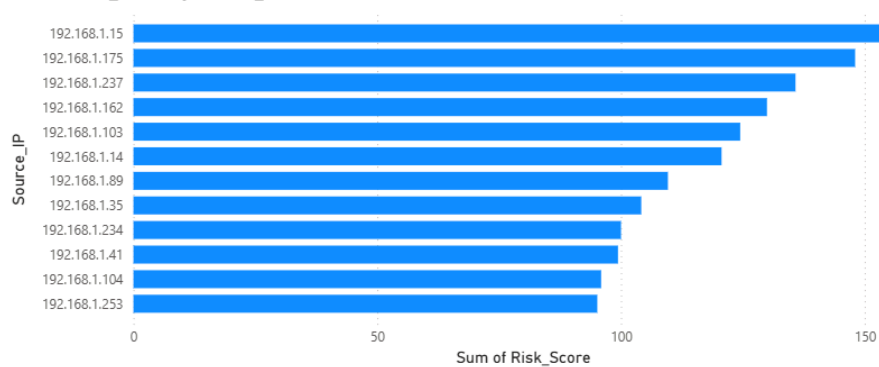
Sum of Login_Attempts Sum of Risk_Score



Count of Threat_Type by Threat_Type



Sum of Risk_Score by Source_IP



Count of Is_Threat by Threat_Type and Threat_Type

